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import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.preprocessing import LabelEncoder

from sklearn.metrics import accuracy_score


data = pd.DataFrame({
    "Age": [25,35,45,28,50,40,30,55,48,32,29,52,41,36,27,60,43,38,31,49],
    "Income": [30000,50000,80000,35000,90000,70000,45000,100000,85000,55000,
               32000,95000,75000,60000,40000,110000,72000,65000,48000,88000],
    "CreditScore": [650,700,750,680,800,720,690,820,780,710,
                    640,810,760,730,660,830,740,700,675,790],
    "LoanAmount": [10000,20000,30000,12000,40000,25000,15000,50000,35000,18000,
                   10000,45000,30000,22000,14000,55000,28000,20000,16000,38000],
    "LoanApproved": ["No","Yes","Yes","No","Yes","Yes","No","Yes","Yes","Yes",
                     "No","Yes","Yes","Yes","No","Yes","Yes","Yes","No","Yes"]
})


X = data.drop("LoanApproved", axis=1)
y = data["LoanApproved"]


encoder = LabelEncoder()
y = encoder.fit_transform(y)


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

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```
model = GaussianNB()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Naive Bayes Bank Loan Prediction")
print("Accuracy:", round(accuracy * 100, 2), "%")

new_customer = [[40, 75000, 730, 25000]]

prediction = model.predict(new_customer)
result = encoder.inverse_transform(prediction)

print("Loan Prediction:", result[0])
```

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... Naive Bayes Bank Loan Prediction
    Accuracy: 100.0 %
    Loan Prediction: Yes
```